AML-3406

AL and ML Capstone Project

Capstone Project Weekly Progress Report

Semester	Fall-2024 Semester 3rd
Course Code	AML-3406
Section	Section 1
Project Title	Car Damage Detection
Group Name	Group B
Student names/Student IDs	Sakshi (C0908000) – TL Bansil Patel (C0912873) Harsh Mohile (C0912872) Meet Patel (C0910378) Rachit Bhatt (C0902810)
Reporting Week	Week 10
Faculty Supervisor	William Pourmajidi

1. Grade the level of collaboration from different aspects between team members: (use: good, medium, below expectation)

Sakshi	Bansil Patel	Harsh Mohile	Meet Patel	Rachit Bhatt
Good	Good	Good	Good	Good

2. Tasks outlined in the previous Weekly Progress Report for this reporting week at the individual level and group level

2.1 Individual tasks (each team member)

Sakshi: Investigate how GitHub Actions can be integrated into the project for automating testing and deployment.

Bansil Patel & Meet Patel: Continue analyzing the model metrics and complete the hyperparameter tuning to optimize performance.

Harsh Mohile & Rachit Bhatt: Demonstrate a fully automated Docker deployment process for multiple environments (e.g., development, staging, production) using GitHub Actions.

2.2 Your team's tasks

- Finalize the analysis of the model's performance metrics and proceed with hyperparameter optimization.
- Continue enhancing the Docker deployment workflow using GitHub Actions and ensure it works seamlessly across different environments.

3. Progress made in Reporting Week at an individual level and group level

3.1 Individual progress

Sakshi: Set up Docker containers and explored Docker Compose for multi-container management.

Bansil Patel & Meet Patel: Evaluated model performance (accuracy, speed) and started hyperparameter tuning on the selected model.

Harsh Mohile & Rachit Bhatt: Demonstrated automated deployment using GitHub Actions and Docker, optimizing windows.

3.2 Your team's progress

- Configured GitHub Actions to automate Docker deployment and set up a CI/CD pipeline.
- Completed model evaluation and refining the selected model with hyperparameter tuning.

4. The areas/tasks you could not make progress and/or complete as scheduled or the difficulties encountered in this reporting week at individual level and group level.

4.1 Individual project blockers

Sakshi: N/A

Bansil Patel: N/A

Meet Patel: N/A

Harsh Mohile & Rachit Bhatt: Faced challenges while setting up CI/CD workflows for multiple branches in GitHub, particularly in aligning Docker deployment with branch specific configurations.

4.2 Your team's project blockers

- While no major blockers were encountered this week, the team experienced some delays in configuring the CI/CD pipeline for multiple branches, particularly when adjusting the Docker deployment scripts.

5. Tasks to be completed in the next week at individual level and group level

5.1 Individual tasks

Sakshi: Investigate how GitHub Actions can be integrated into the project for automating testing and deployment.

Bansil Patel & Meet Patel: Continue analyzing the model metrics and complete the hyperparameter tuning to optimize performance.

Harsh Mohile & Rachit Bhatt: Researching Deployment UI and Backend.

5.2 Your team's tasks

- Finalize the analysis of the model's performance metrics and proceed with hyperparameter optimization.
- Researching Deployment UI and Backend.

6. Include the tasks from your sprint planning (GitHub/ZenHub) for the present period.

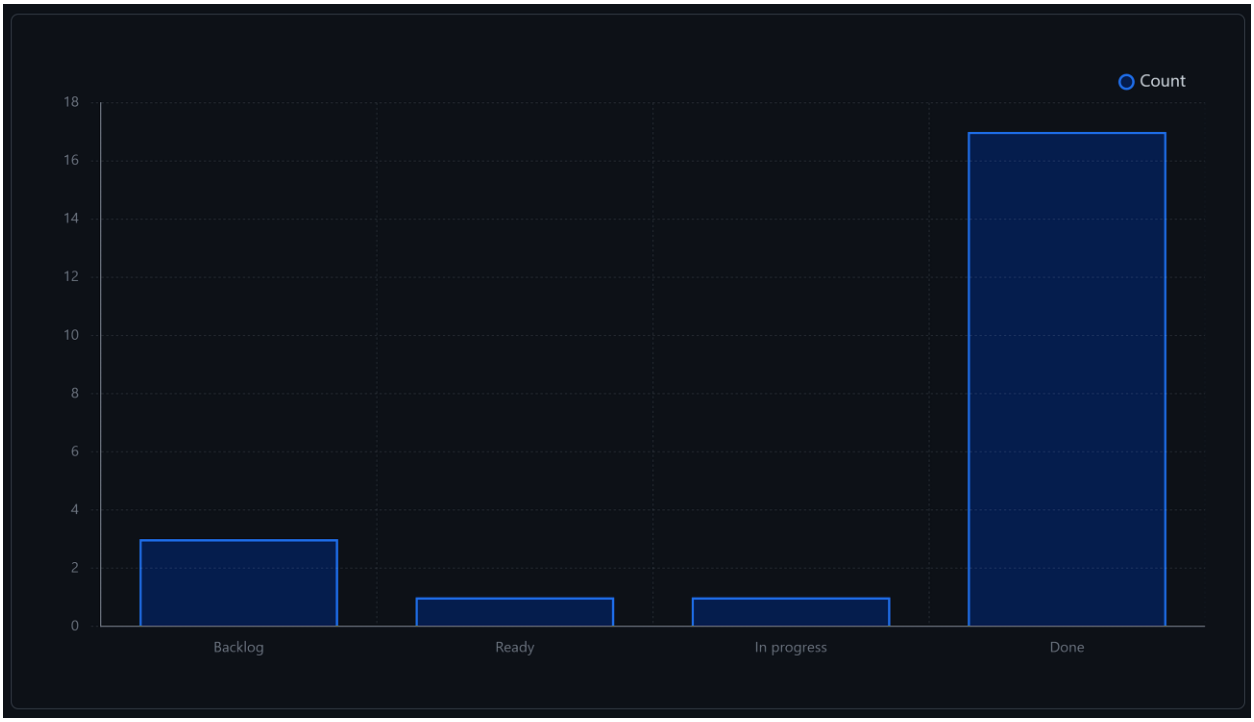

Figure 1: Weekly Activities

- In this week, there were a lot of issues resolved from the backlog, contributing to development.

7. Include charts/graphs (e.g., burn down charts) from your project management tool (Github/Zenhub) that shows your progress for the period of this report.

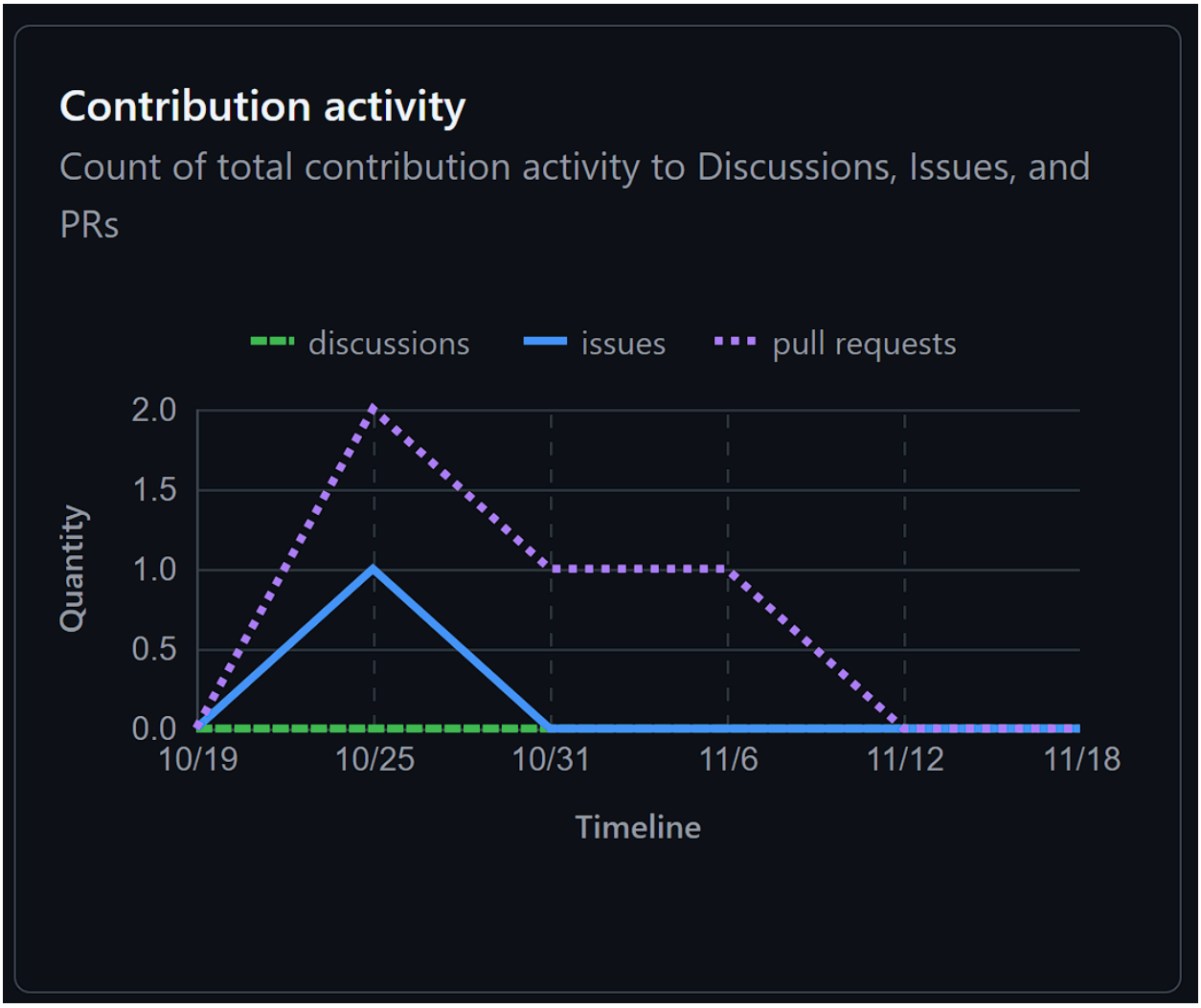

Figure 2: Contributors in the Week

- This week, we had a few PRs and issues resolved as compared to previous week, demonstrating team hard work and dedication.

8. **Include a note and address to your project GitHub with list of codes uploaded/updated on GitHub in this reporting week.**

GitHub: https://www.github.com/rachit-bhatt/Capstone-Project